



Security

ACE Team

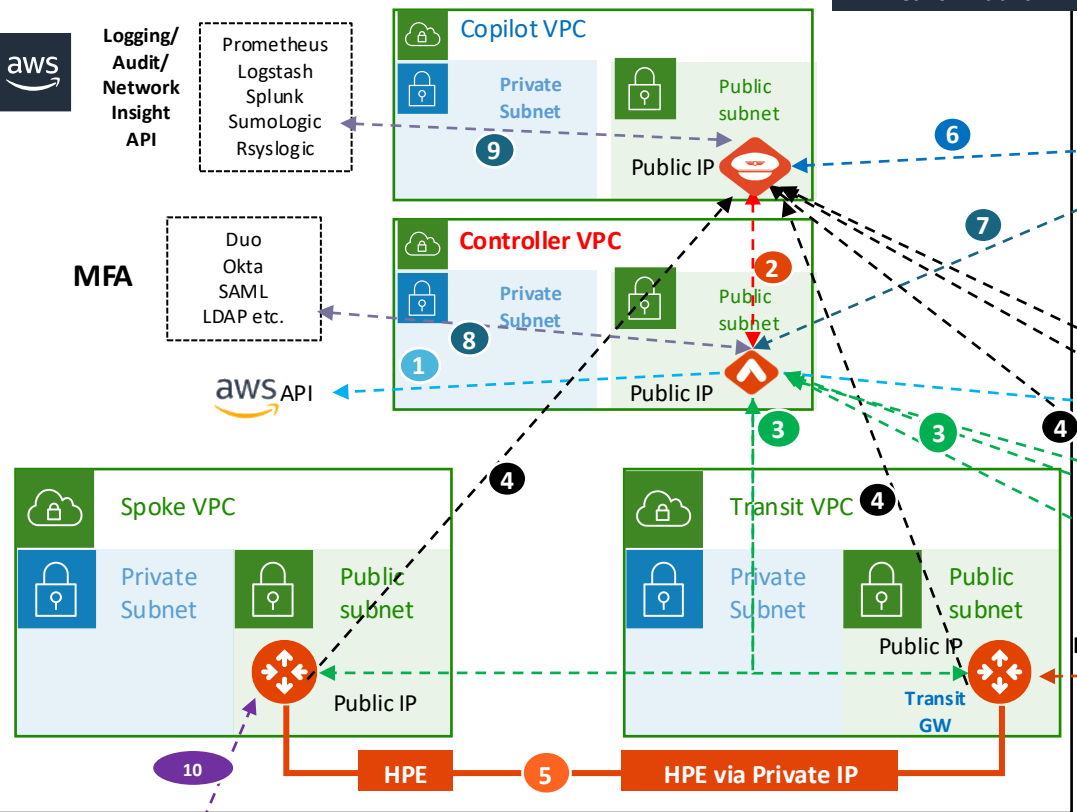
Agenda

- Aviatrix Security Features Overview
- Securing Aviatrix Platform
- Aviatrix Cloud Firewall
- Public Subnet Filtering Gateway



Built-in Security of the Aviatrix Platform

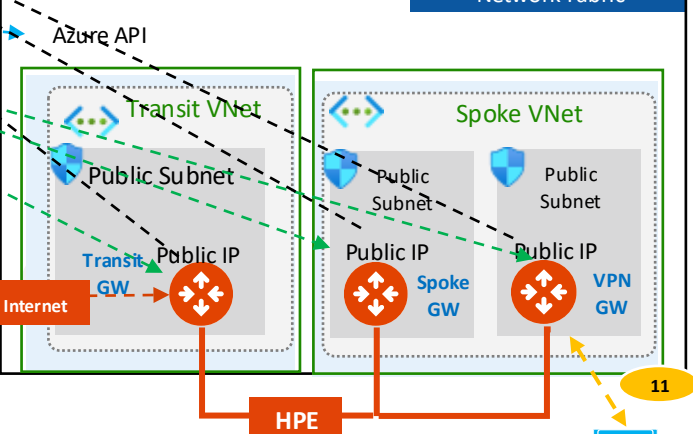
AWS Cloud



Traffic inside AWS Network Fabric

- ### Traffic Pattern
1. Controller to CSP API
 2. Controller with Copilot
 3. Controller to GW management traffic
 4. Gateway to Copilot (Syslog, Netflow etc)
 5. Encrypted data transfer
 6. Copilot access locked to customer IP
 7. Controller access locked to customer IP
 8. Controller to MFA
 9. Copilot to Customers Network Insight API or Logging locations
 10. Aviatrix Gateway to 3rd Party devices
 11. Remote user to Aviatrix VPN gateway

Azure Cloud



Traffic inside Azure Network Fabric

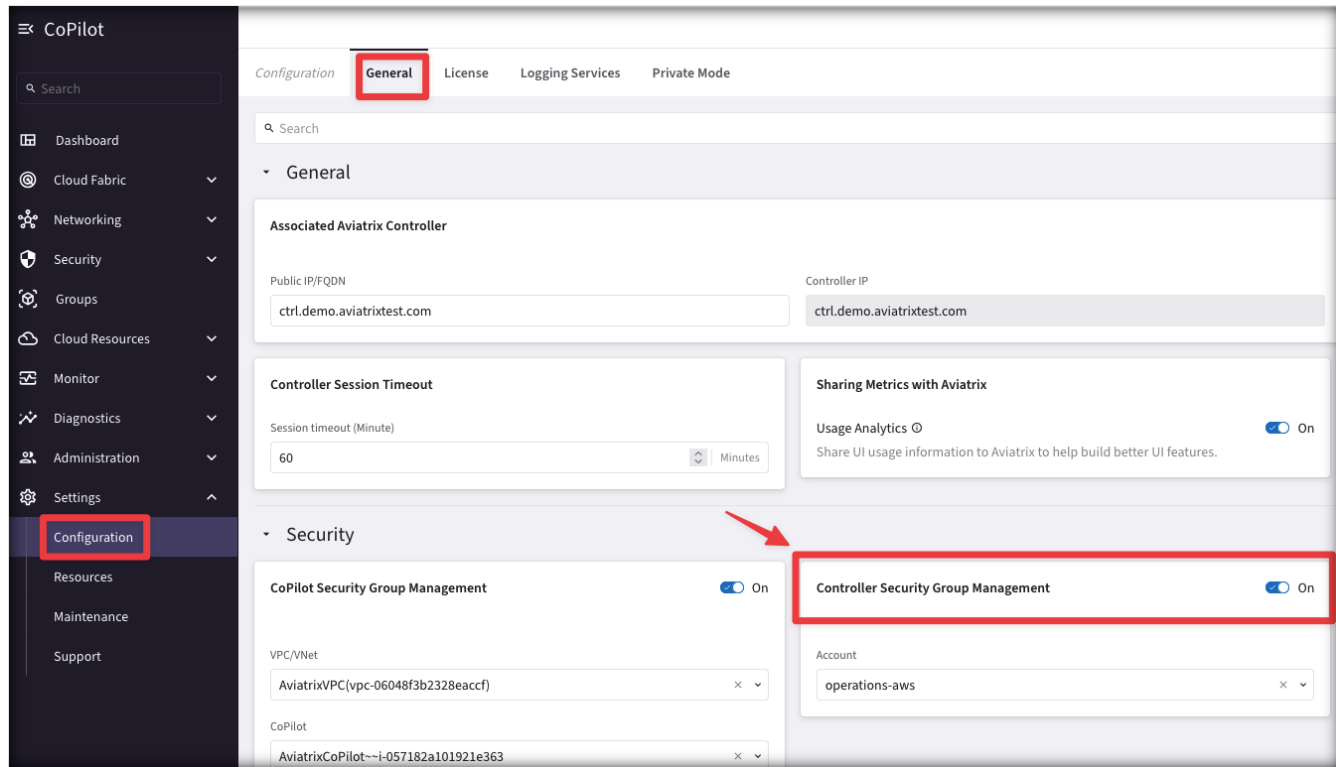
On Prem DC/
Branch Office/
B2B Partner

Remote User

Controller Security Group Management (part.1)

- You can use the **Controller Security Group Management** feature to automatically manage the Controller instance's inbound rules from gateways.
- When enabled (**default**), each time you deploy an Aviatrix gateway, a rule will be automatically added to the Controller instance's inbound rule to allow the gateway to reach the Controller. Only TCP port 443 needs to be opened for inbound traffic to the Controller. Gateways launched from the Controller use its public IP address to communicate back to the Controller.
- After the Controller Security Group Management feature is enabled, you can edit the security rules that are outside gateways public IP addresses to limit the source address range. When specifying the custom IP addresses to allow access, you must include your own public IP address.

Controller Security Group Management (part.2)



The screenshot displays the Aviatrix CoPilot Configuration interface. The left-hand navigation menu has 'Configuration' selected. The main panel shows the 'General' tab under the 'Configuration' section. Key settings visible include:

- Associated Aviatrix Controller:** Public IP/FQDN and Controller IP are both set to `ctrl.demo.aviatrixtest.com`.
- Controller Session Timeout:** Set to 60 minutes.
- Sharing Metrics with Aviatrix:** Usage Analytics is turned On.
- Security Section:**
 - CoPilot Security Group Management:** Enabled (On).
 - Controller Security Group Management:** Enabled (On). This section is highlighted with a red box, and a red arrow points to it from the left.

- You can enable Controller Security Group Management in CoPilot from **Settings > Configuration > General**

CoPilot Security Group Management (part.1)

- When **CoPilot Security Group Management** is enabled (**default**), the Controller creates a security group for the specified CoPilot virtual machine to manage its inbound security-group rules.

The feature adds gateway IP rules to customer-attached CoPilot security groups as well as CoPilot-created security groups. CoPilot comes with a base security group when it is first launched.

The Controller adds rules to the security group for each gateway IP for the following:

- **UDP port 5000** (default) — Enable Syslog for CoPilot Egress FQDN (Legacy) & Audit Data (from each gateway). Gateways send remote syslog data to CoPilot.
- **TCP port 5000** (default, if using Private Mode) — Enable Syslog for CoPilot Egress FQDN & Audit Data (from each gateway). Gateways send remote syslog data to CoPilot.
- **UDP port 31283** (default, port is configurable) — Enable NetFlow for CoPilot FlowIQ Data (from each gateway). Gateways send NetFlow to CoPilot.

The Controller adds the above rules for:

- New gateways launched from the Controller after the feature is enabled.
- Existing gateways launched from the Controller before the feature was enabled.

CoPilot Security Group Management (part.2)

The screenshot displays the Aviatrix CoPilot Configuration interface. The left-hand navigation menu has 'Configuration' selected. The main content area shows the 'General' configuration tab. Key settings include the 'Associated Aviatrix Controller' (Public IP/FQDN and Controller IP both set to ctrl.demo.aviatrixtest.com), 'Controller Session Timeout' (60 minutes), and 'Sharing Metrics with Aviatrix' (Usage Analytics is On). In the 'Security' section, 'CoPilot Security Group Management' is highlighted with a red box and its toggle is turned On. Below this, 'Controller Security Group Management' is also turned On. Further down, there are dropdown menus for VPC/VNet (AviatrixVPC) and CoPilot (AviatrixCoPilot).

- You can enable CoPilot Security Group Management in CoPilot from **Settings > Configuration > General**



Securing the Platform with Cloud Native Load Balancers

Problem Statement

- Enterprise concerns around putting Aviatrix Controller with a public IP in a Public subnet
- Enterprises need tighter security and availability
- What are the options?
 1. Limit access using cloud native L4 stateful firewalls such as:
 - AWS Security Groups
 - Azure Network Security Groups
 - GCP Firewall Rules
 2. Deploy a third-party Firewall in front of controller
 3. Deploy an Application (L7) Load Balancer in front of Aviatrix Controller

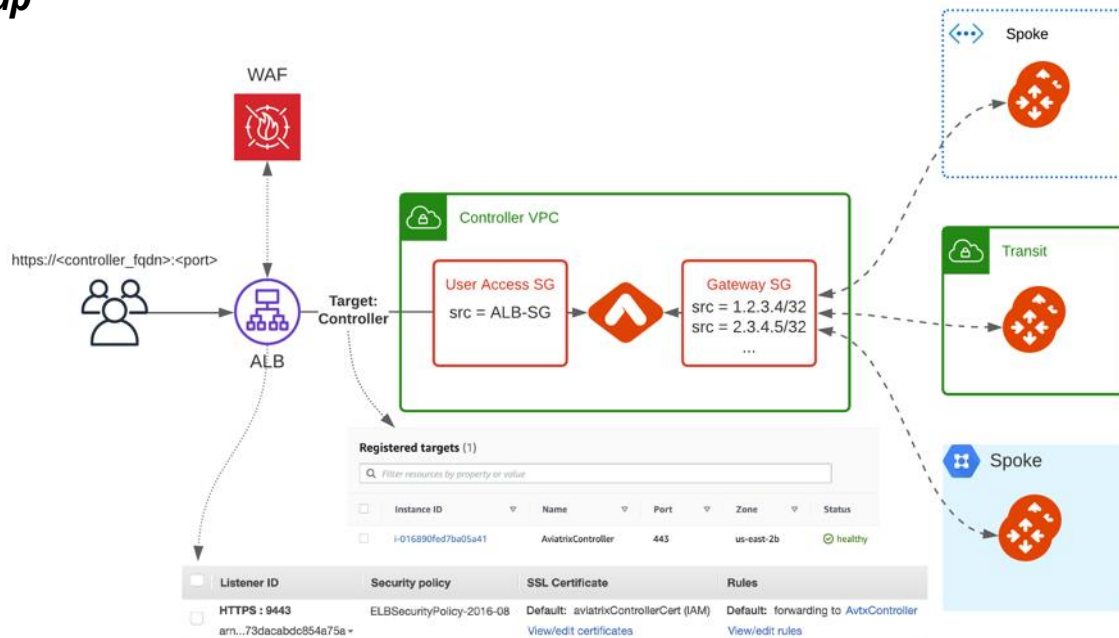
- Verify that the **Controller Security Group Management** feature is NOT disabled.

This feature allows access to the Controller EIP from Aviatrix Gateways, solely

- Create a new internet facing ALB
- Modify main Controller Security Group to only allow access from the ALB Security Group

- Enable WAF on the ALB with AWS Managed Rules

- Adjust ALB idle timeout, modify rulesets
- Modify ALB Security Group to only allow access from the admin user IP





Aviatrix Cloud Firewall

Private workloads need internet access

- SaaS integration

- Patching



- Updates



Understanding the Pain

Improve Security and Lower Cloud Costs

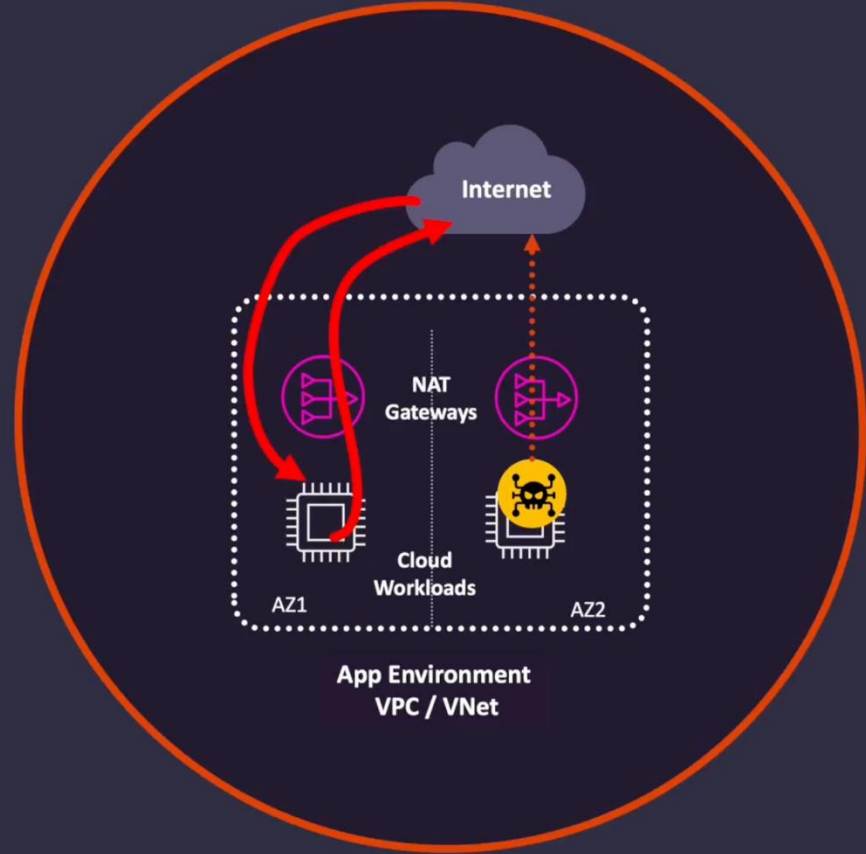


- **Business Pain**

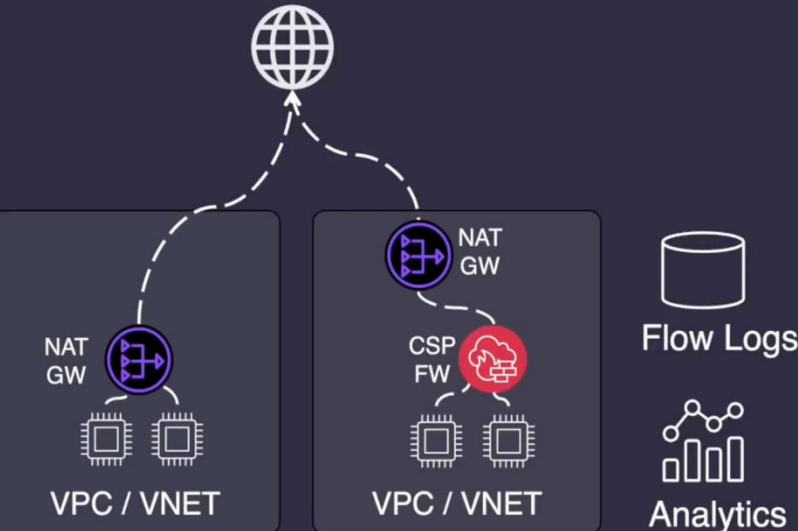
- Excessive Cloud Costs
- Lack of Compliance & Governance
- Risk to Business-Critical Workloads
- Regulatory Fines and Penalties
- Brand Health and Customer Trust

- **Technical Pain**

- No Policy Enforcement
- Slow Troubleshooting and Forensics
- Identifying Noisy Workloads
- Support Distributed Deployments
- Advanced Inspection Capabilities



Two Common Paths



1. Distributed Cloud Provider Services

- Expensive: High data-processing costs
- Zero / Weak Security
- Poor Visibility
 - Some visibility with a lot of tools
- Log storage and analytics costs
- No centralized intelligence
- Not multi-cloud capable

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CLOUD SECURITY

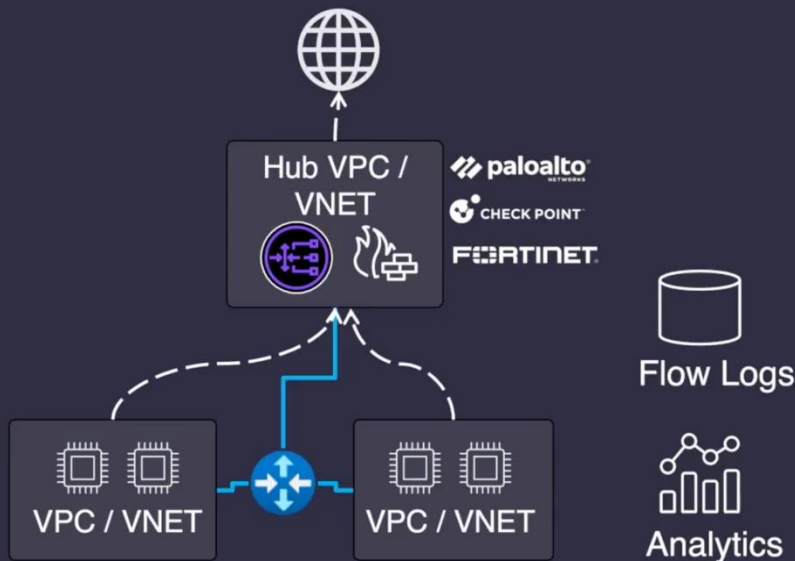
CyberRatings.org Announces Test Results for Cloud Service Provider Native Firewalls

Protection ranged from 0.38% to 50.57% for security effectiveness.

Two Common Paths

2. Central Virtualized Appliances

- Very Expensive
- Not built for cloud: operational complexity
- No support for Island VPCs / VNets
- Requires Overly Complex Routing Architecture
- Security Hub Connectivity dependent
- No centralized network and security intelligence
- Additional troubleshooting issues
- Not multi-cloud deployable



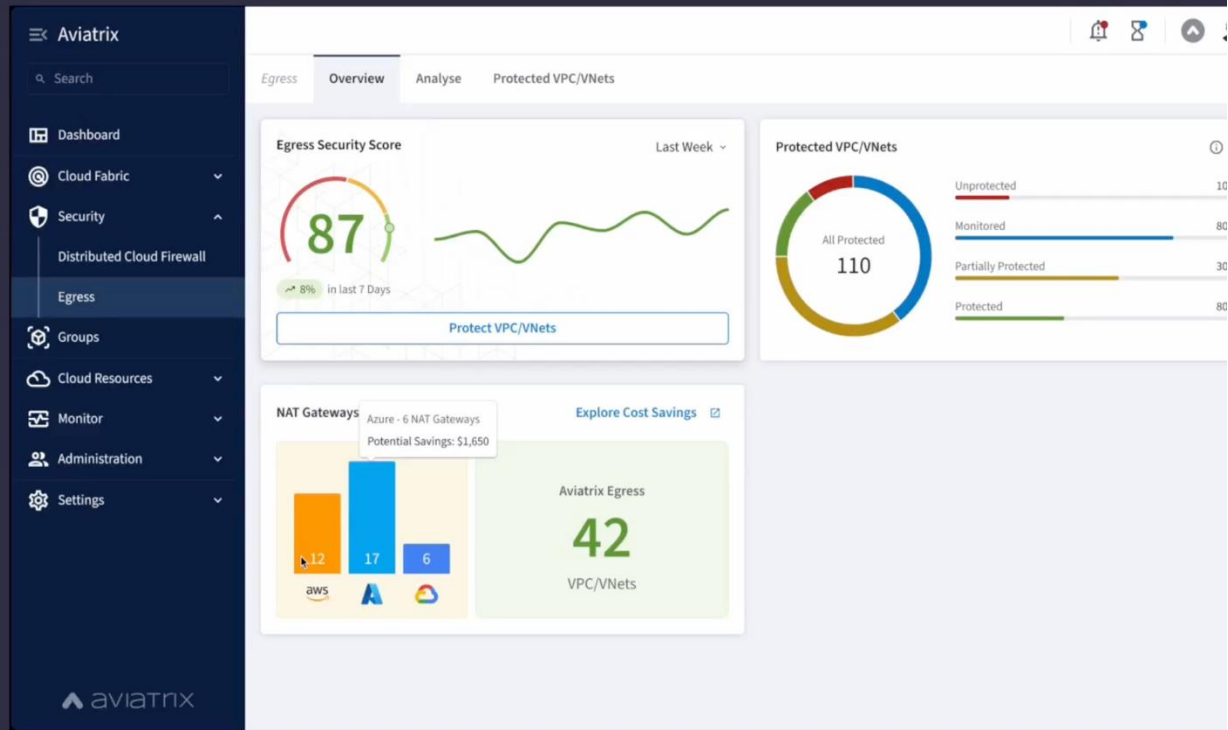
Aviatrix Cloud Firewall

What it is:

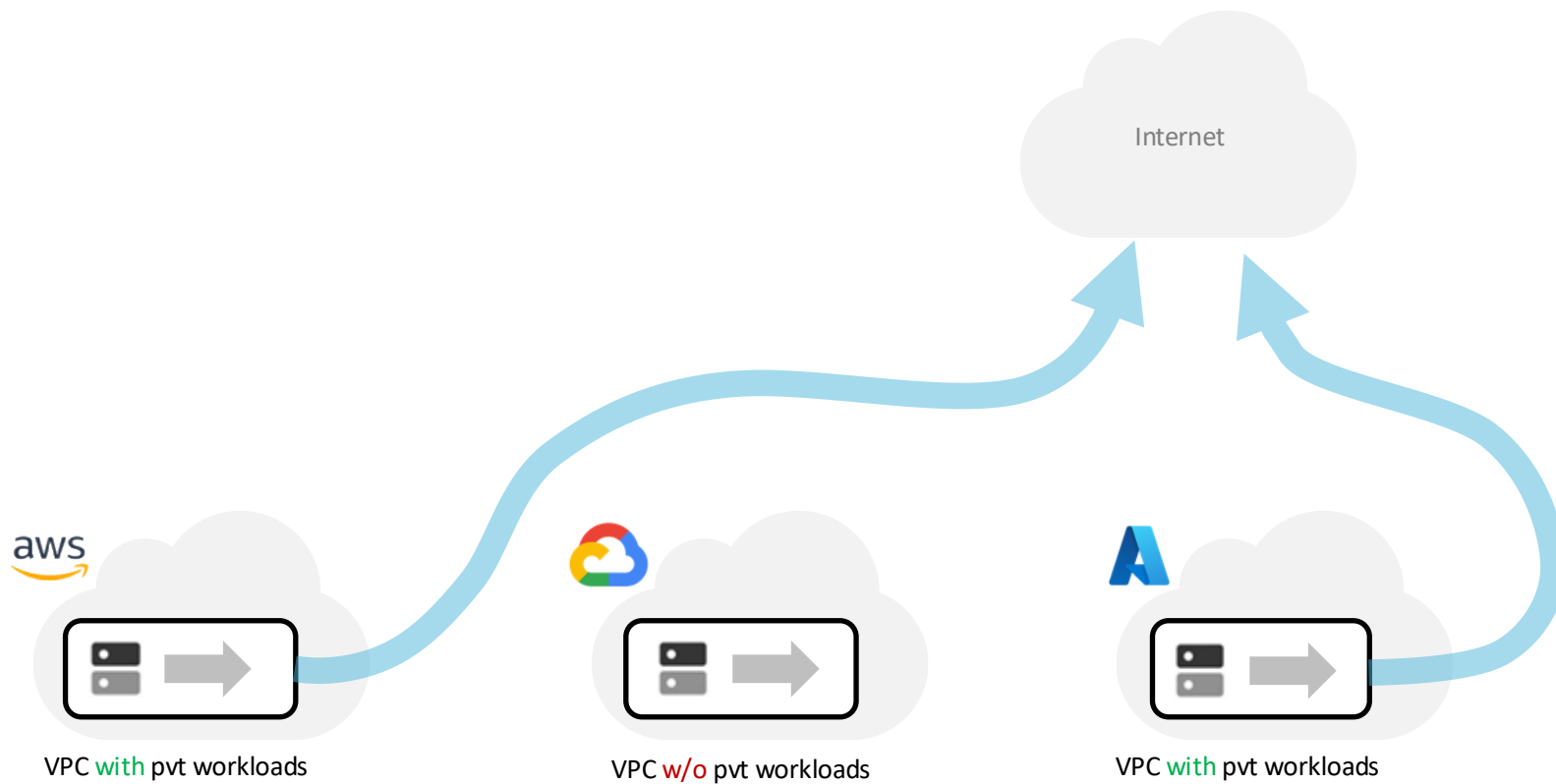
- Central Policy Management & Observability
- Distributed Enforcement: at the workload

What you get:

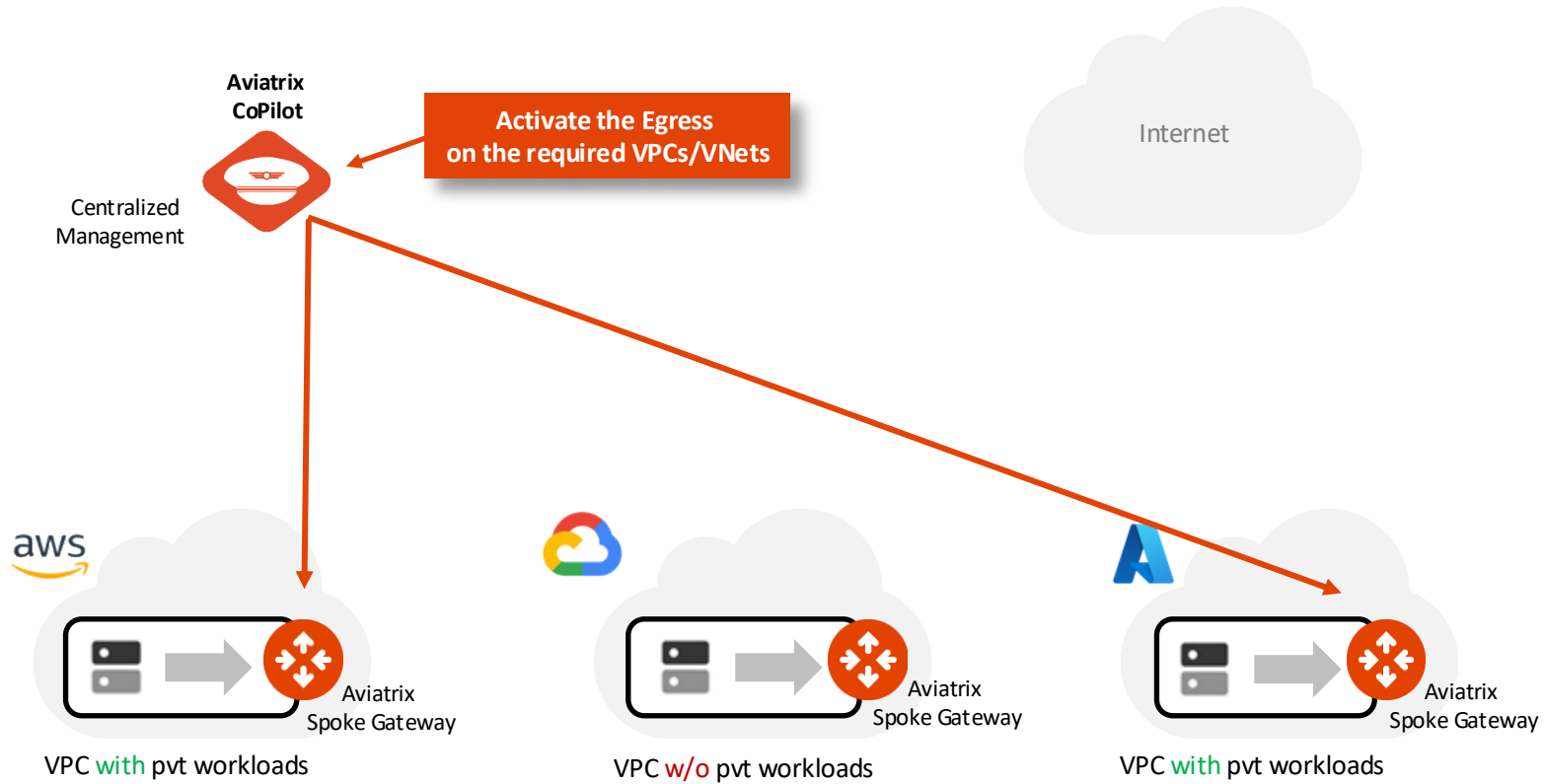
- Secure Networking that's:
 - Agile,
 - Reduces Costs & Complexity
 - Increases Visibility



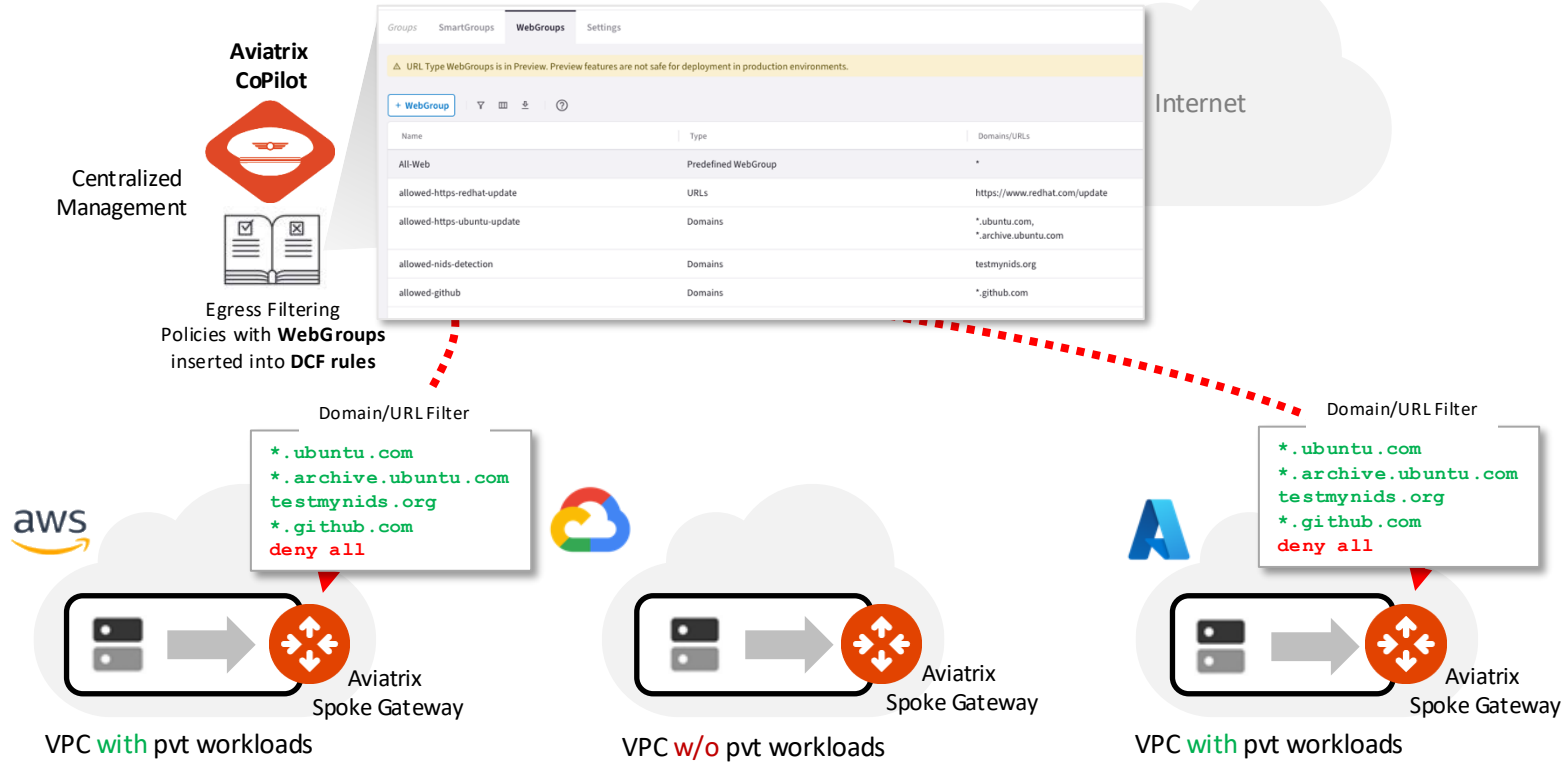
Aviatrix Cloud Firewall



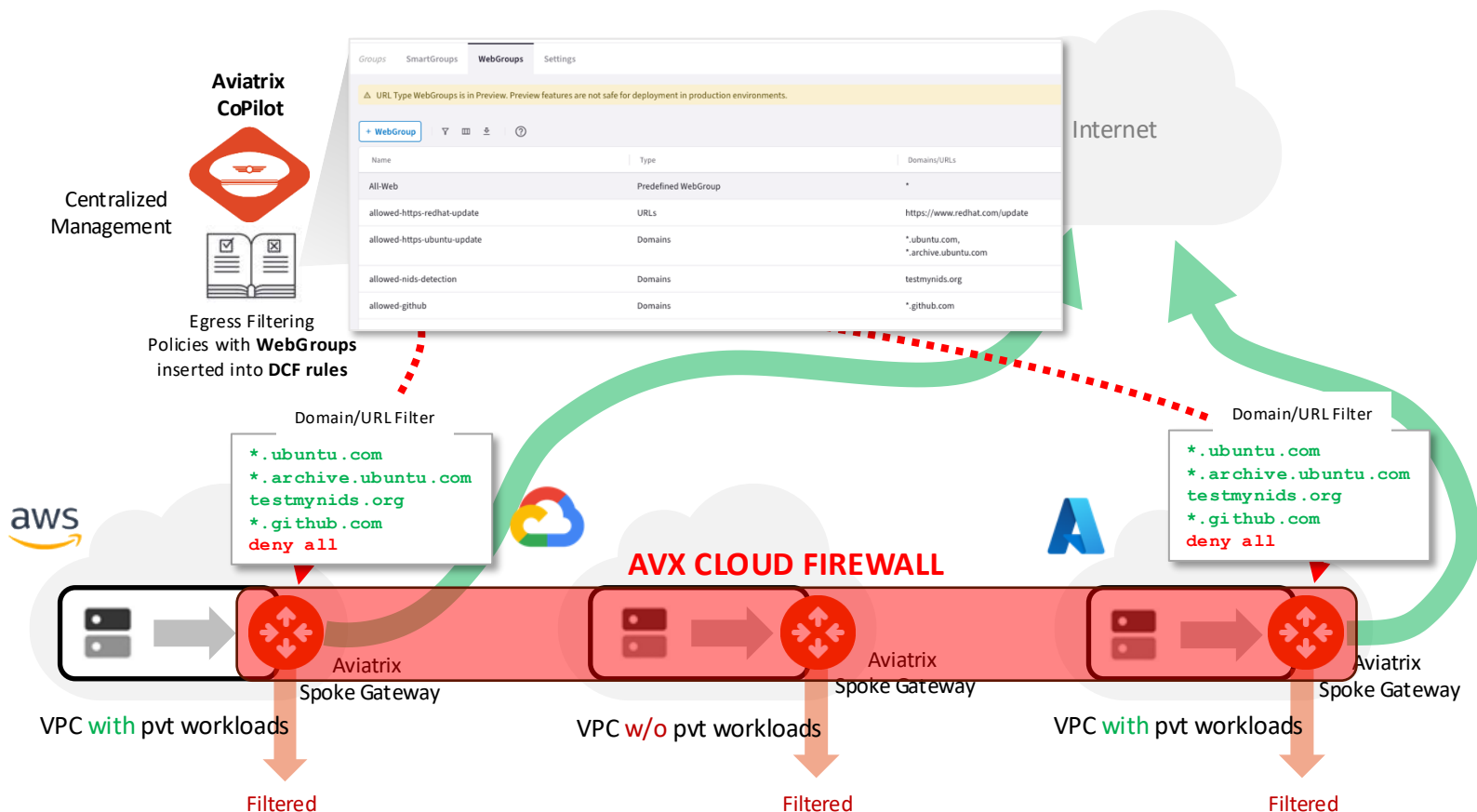
Aviatrix Cloud Firewall



Aviatrix Cloud Firewall



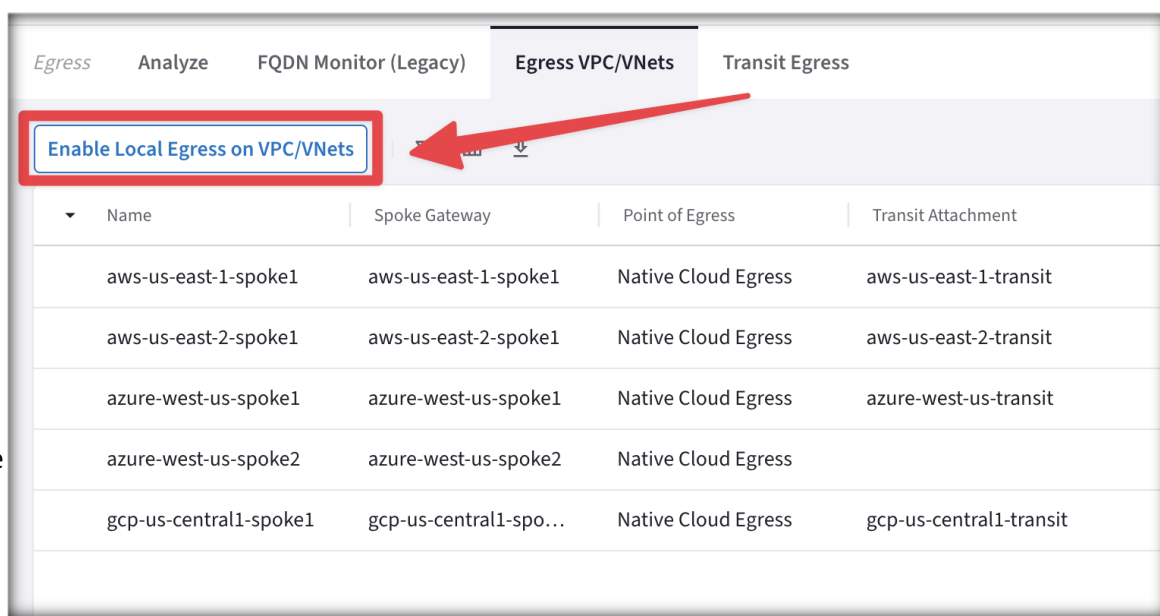
Aviatrix Cloud Firewall



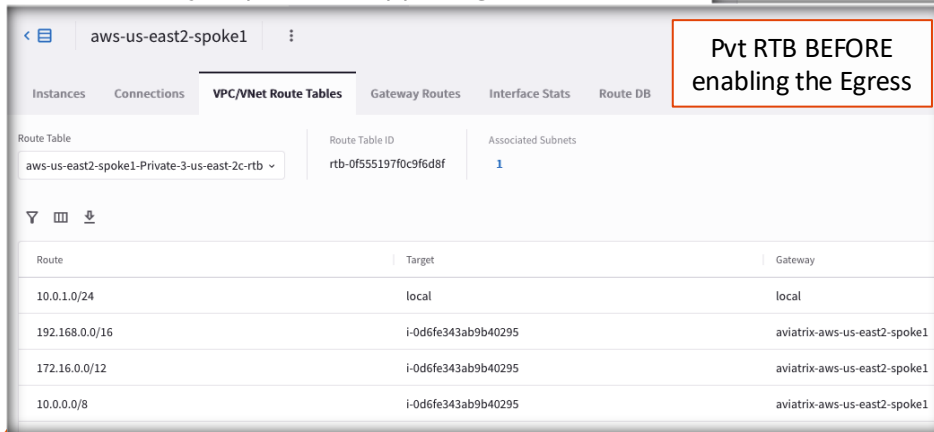
- The Aviatrix Cloud Firewall can be extended also to the Edge

Enabling Egress

- Adding Egress Control on VPC/VNet changes the default route on VPC/VNet to point to the Spoke Gateway and enables **SNAT**.
- In addition to the **Local route**, the **three RFC1918 routes**, also a **default route** will be injected.
- CAVEAT: Egress Control also requires additional resources on the Spoke Gateway (i.e. scale up the VM size). Before enabling Egress Control on Spoke Gateways, ensure that you have created the additional CPU resources on the Spoke Gateway required to support Egress Control.

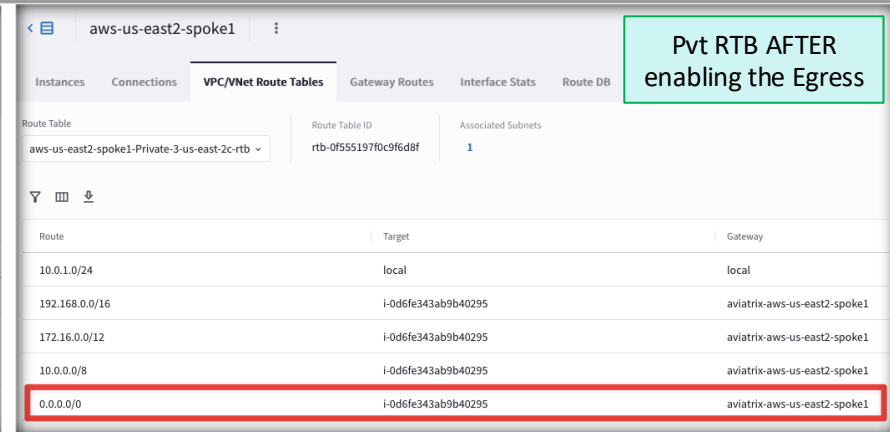


Name	Spoke Gateway	Point of Egress	Transit Attachment
aws-us-east-1-spoke1	aws-us-east-1-spoke1	Native Cloud Egress	aws-us-east-1-transit
aws-us-east-2-spoke1	aws-us-east-2-spoke1	Native Cloud Egress	aws-us-east-2-transit
azure-west-us-spoke1	azure-west-us-spoke1	Native Cloud Egress	azure-west-us-transit
azure-west-us-spoke2	azure-west-us-spoke2	Native Cloud Egress	
gcp-us-central1-spoke1	gcp-us-central1-spo...	Native Cloud Egress	gcp-us-central1-transit



Pvt RTB BEFORE enabling the Egress

Route	Target	Gateway
10.0.1.0/24	local	local
192.168.0.0/16	i-0d6fe343ab9b40295	aviatrix-aws-us-east2-spoke1
172.16.0.0/12	i-0d6fe343ab9b40295	aviatrix-aws-us-east2-spoke1
10.0.0.0/8	i-0d6fe343ab9b40295	aviatrix-aws-us-east2-spoke1



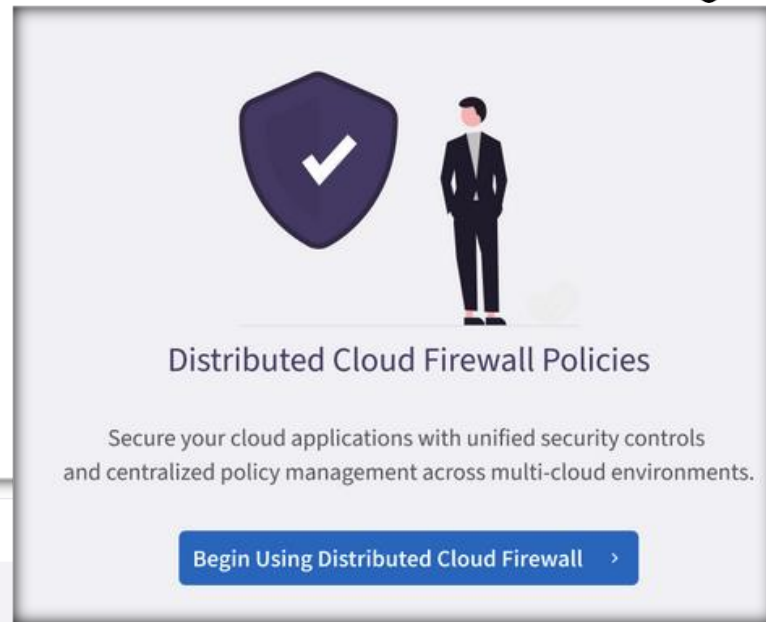
Pvt RTB AFTER enabling the Egress

Route	Target	Gateway
10.0.1.0/24	local	local
192.168.0.0/16	i-0d6fe343ab9b40295	aviatrix-aws-us-east2-spoke1
172.16.0.0/12	i-0d6fe343ab9b40295	aviatrix-aws-us-east2-spoke1
10.0.0.0/8	i-0d6fe343ab9b40295	aviatrix-aws-us-east2-spoke1
0.0.0.0/0	i-0d6fe343ab9b40295	aviatrix-aws-us-east2-spoke1

Enabling Distributed Cloud Firewall (DCF)

- When Distributed Cloud Firewall is enabled in Aviatrix CoPilot, a **Default Action Rule** is automatically created in the Post Rules Policy List (System). This global rule governs the default behavior for traffic that does not match any user-defined rules.

CAVEAT: This is a system rule used to enforce *zero trust principles* by controlling how traffic is handled in the absence of explicit rules.



Distributed Cloud Firewall

Policies Monitor Detected Intrusions Settings

Ruleset **Post Rules Policy List (System)** [Manage Rulesets](#)

[+ Rule](#) | [Actions](#) | [Filter](#) | [Grid](#) | [Download](#) | [Help](#)

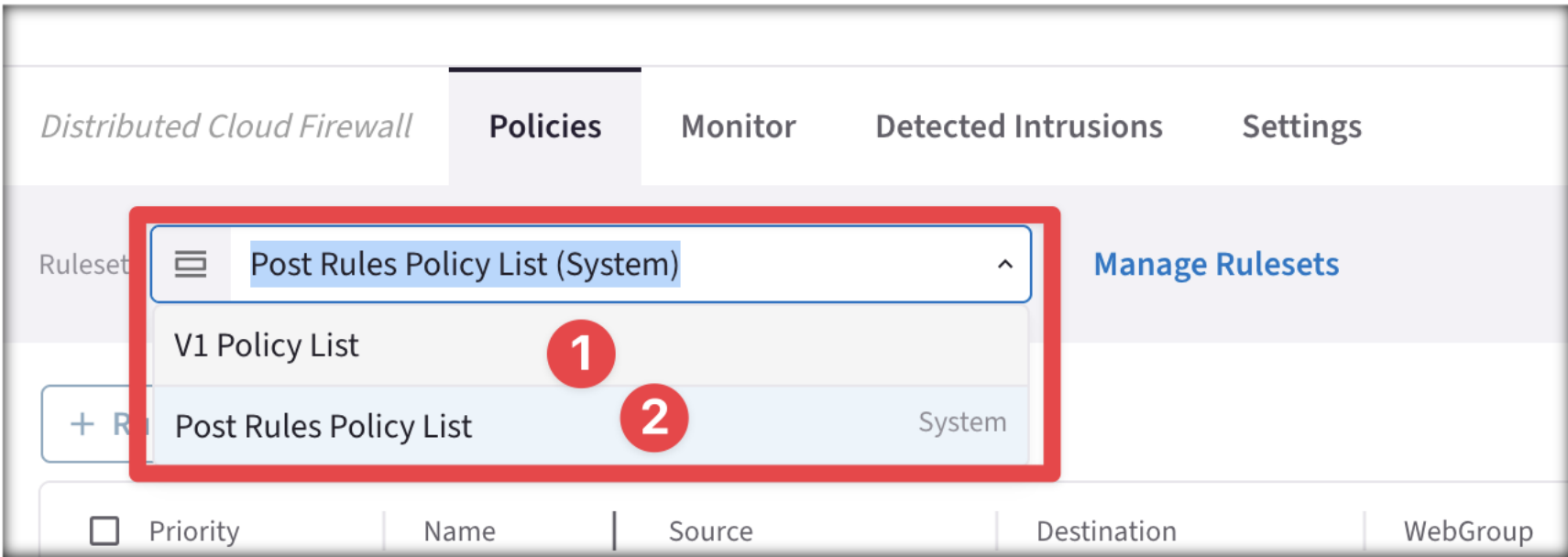
Search Modified View

Priority	Name	Source	Destination	WebGroup	Protocol	Ports	Action
☐ 214748...	Default Action Rule	Anywhere (0.0.0.0/0)	Anywhere (0.0.0.0/0)		Any		Permit

Managing Distributed Cloud Firewall Rulesets

- Two rulesets are initially available when you start using the Rulesets feature:
 - **Post Rules Policy List:** System-based ruleset that contains the Default Action Rule. You cannot delete this ruleset or the Default Action Rule. This ruleset is executed last in the ruleset list.
 - **V1 Policy List:** Ruleset added by Aviatrix. You can add rules to this ruleset, but you cannot delete it.

CAVEAT: After enabling the Distributed Cloud firewall, configure the Ruleset to the **V1 Policy List** and begin populating it with your custom rules.



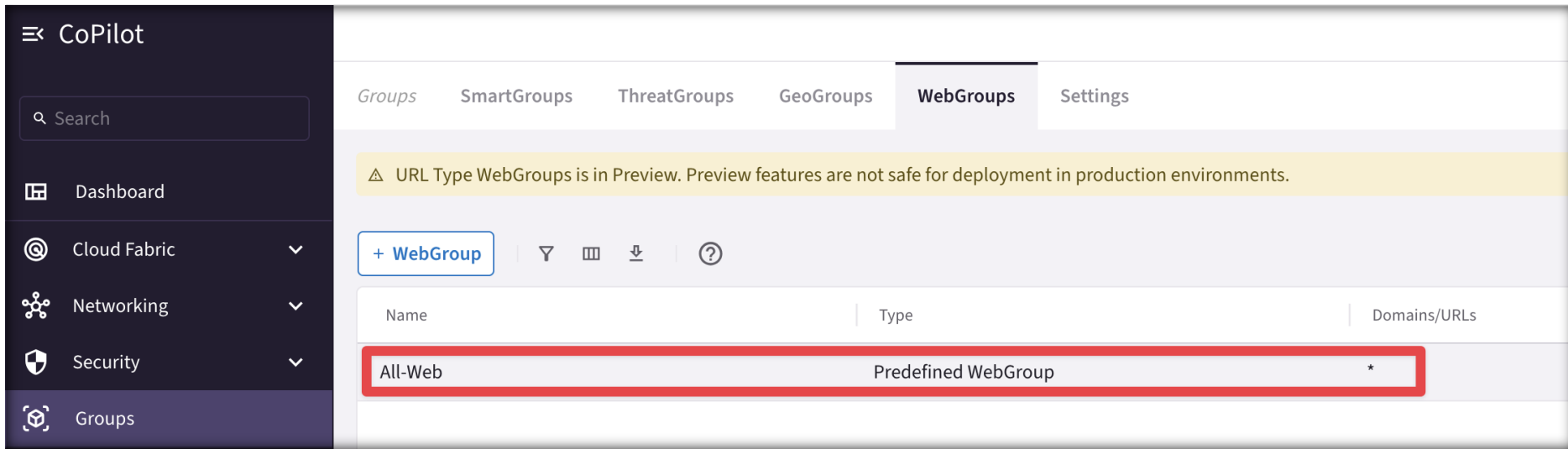
The screenshot shows the 'Distributed Cloud Firewall' interface with the 'Policies' tab selected. A dropdown menu is open, showing 'Post Rules Policy List (System)' and 'V1 Policy List'. A red box highlights the dropdown and the 'Post Rules Policy List' entry in the table below. Red circles with numbers 1 and 2 point to the 'V1 Policy List' and 'Post Rules Policy List' entries respectively.

Ruleset	Manage Rulesets
Post Rules Policy List (System)	
V1 Policy List	
Post Rules Policy List	System

Below the table, there are columns for Priority, Name, Source, Destination, and WebGroup.

Predefined WebGroup: All-Web

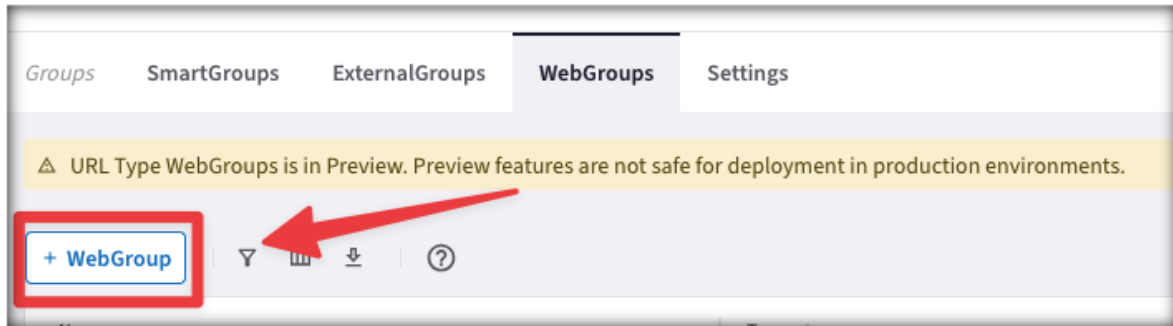
- When you navigate to **CoPilot > Groups**, a predefined WebGroup, *All-Web*, has already been created for you.
- This is an "allow-all" WebGroup that you must select in a Distributed Cloud Firewall rule if you do not want to limit the Internet-bound traffic for that rule, but you still want to log the FQDNs that are being accessed.

The screenshot shows the Aviatrix CoPilot web interface. On the left is a dark sidebar with a hamburger menu icon and the text "CoPilot". Below this is a search bar and a list of navigation items: "Dashboard", "Cloud Fabric", "Networking", "Security", and "Groups" (which is highlighted). The main content area has tabs for "Groups", "SmartGroups", "ThreatGroups", "GeoGroups", "WebGroups" (selected), and "Settings". A yellow warning banner at the top of the main area states: "URL Type WebGroups is in Preview. Preview features are not safe for deployment in production environments." Below the banner is a toolbar with a "+ WebGroup" button and icons for filter, table view, download, and help. A table below the toolbar lists WebGroups. The first row is highlighted with a red border and contains the text "All-Web", "Predefined WebGroup", and an asterisk "*".

Name	Type	Domains/URLs
All-Web	Predefined WebGroup	*

WebGroup Creation

- **WebGroups** are groupings of domains and URLs, inserted into Distributed Cloud Firewall rules, that filter (and provide security to) Internet-bound traffic.
- In addition to the predefined WebGroup **All-Web**, you can also create two kind of custom WebGroups:
 1. **URLs WebGroup**: for HTTP/HTTPS and for other protocols, but you need to define the full Path.
 - CAVEAT: TLS Decryption must be turned on when URLs-based WebGroups are used.
 2. **Domains WebGroup**: for HTTP and HTTPS traffic (wild cards are supported – i.e. partial names).



Create WebGroup

Name
FTP-to-Example.com

Type
☐ Domains
 ☒ URLs

Domains/URLs
ftp://ftp.example.com/directory/ x

Cancel Save

Create WebGroup

Name
Apt-get-Commands

Type
☒ Domains
 ☐ URLs

Domains/URLs
*ubuntu.com x

Cancel Save

Monitor

- On the **FQDN Monitor (Legacy)** section you can retrieve all the logs and therefore distinguish the domains that should be permitted from those ones that should be denied.

Egress Analyze **FQDN Monitor (Legacy)** Egress VPC/VNets Transit Egress

Filters

Time Period: Last 24 Hours Start: Apr 03, 2025 12:00 PM End: Now VPC/VNets: accounting-aws-spoke-dev

Timestamp	Source IP	VPC/VNet	Domain	Port	Rule Match
Apr 4, 2025 11:50 AM	10.1.2.5	accounting-aws-spoke-dev	ssm.us-east-1.amazonaws.com	443	Matched
Apr 4, 2025 11:21 AM	10.1.2.5	accounting-aws-spoke-dev	ssm.us-east-1.amazonaws.com	443	Matched
Apr 4, 2025 11:11 AM	10.1.2.5	accounting-aws-spoke-dev	api.snapcraft.io	443	Matched
Apr 4, 2025 11:10 AM	10.1.2.5	accounting-aws-spoke-dev	api.snapcraft.io	443	Matched
Apr 4, 2025 11:10 AM	10.1.2.5	accounting-aws-spoke-dev	api.snapcraft.io	443	Matched
Apr 4, 2025 11:10 AM	10.1.2.5	accounting-aws-spoke-dev	api.snapcraft.io	443	Matched
Apr 4, 2025 11:10 AM	10.1.2.5	accounting-aws-spoke-dev	api.snapcraft.io	443	Matched
Apr 4, 2025 11:10 AM	10.1.2.5	accounting-aws-spoke-dev	api.snapcraft.io	443	Matched
Apr 4, 2025 10:53 AM	10.1.2.5	accounting-aws-spoke-dev	ssm.us-east-1.amazonaws.com	443	Matched
Apr 4, 2025 10:28 AM	10.1.2.5	accounting-aws-spoke-dev	ssm.us-east-1.amazonaws.com	443	Matched
Apr 4, 2025 9:58 AM	10.1.2.5	accounting-aws-spoke-dev	ssm.us-east-1.amazonaws.com	443	Matched
Apr 4, 2025 9:31 AM	10.1.2.5	accounting-aws-spoke-dev	ssm.us-east-1.amazonaws.com	443	Matched
Apr 4, 2025 9:02 AM	10.1.2.5	accounting-aws-spoke-dev	ssm.us-east-1.amazonaws.com	443	Matched
Apr 4, 2025 8:32 AM	10.1.2.5	accounting-aws-spoke-dev	ssm.us-east-1.amazonaws.com	443	Matched
Apr 4, 2025 8:06 AM	10.1.2.5	accounting-aws-spoke-dev	ssm.us-east-1.amazonaws.com	443	Matched

Top Rules Hit

www.wikipedia.com (80)	3
www.football.com (80)	3
www.espn.com (80)	3
www.aviatrix.com (80)	3
us-east-2.ec2.archive.ubuntu.com (80)	3
security.ubuntu.com (80)	1
esm.ubuntu.com (443)	1

Allowed

Allowed

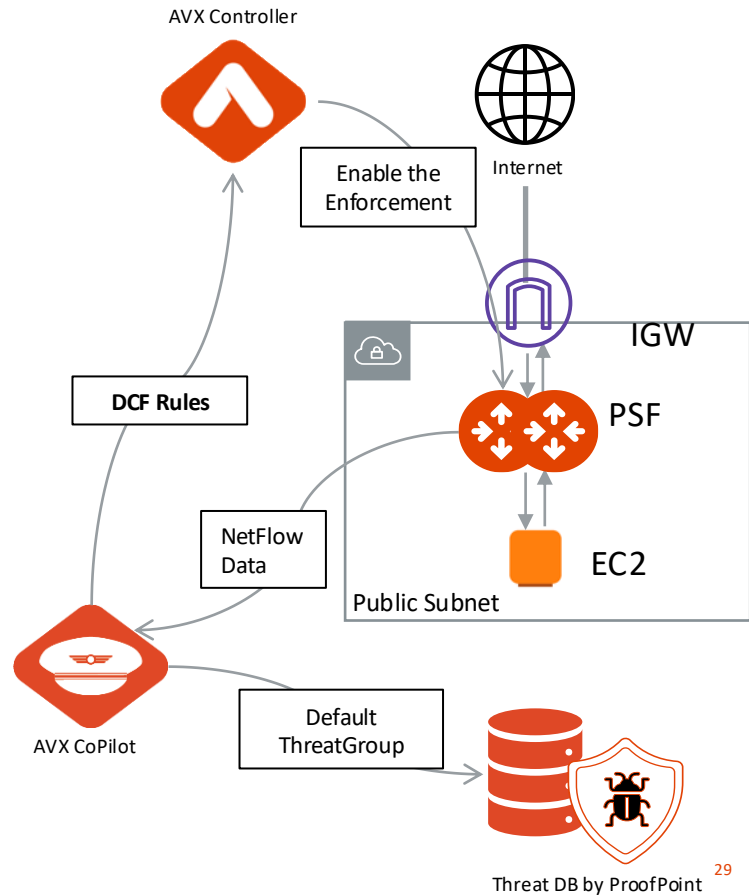
Allowed



Aviatrix PSF GW(aka Public Subnet
Filtering Gateway)

Aviatrix Public Subnet Filtering Gateways (PSF GWs)

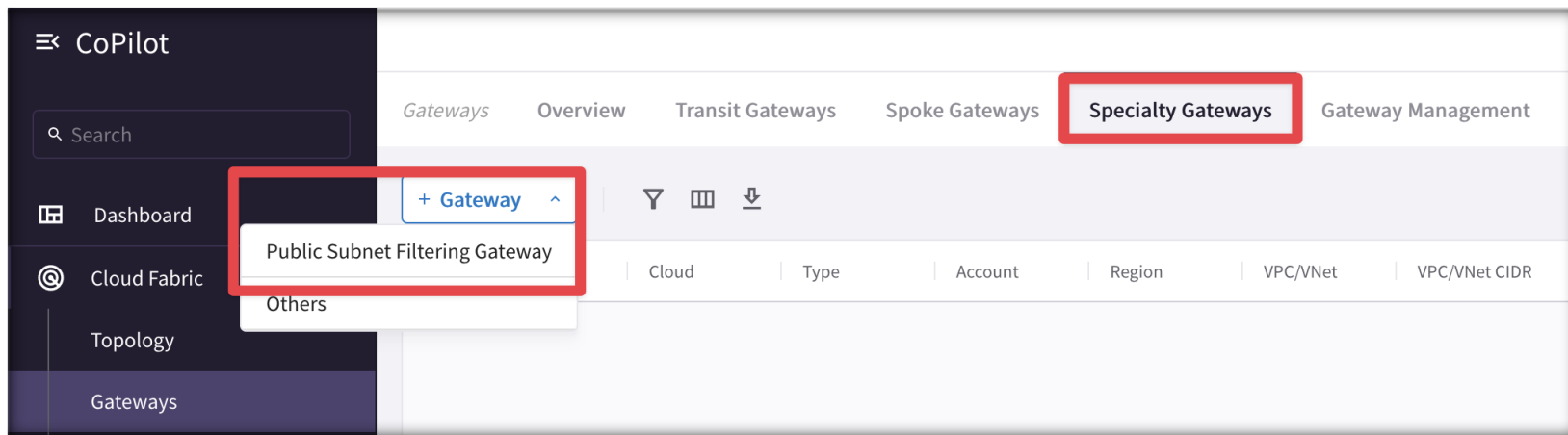
- **Public Subnet Filtering Gateways (PSF** gateways) provide ingress and egress security for **AWS** public subnets where instances have public IP addresses.
- After the Public Subnet Filtering (PSF) gateway is launched, you can apply also DCF (Distributed Cloud Firewall) rules – *enforcement must be enabled*.
- The PSF Gateway acts as a **standalone Gateway** (it's neither a Spoke nor a Transit).
- Leverage the **Default ThreatGroup** (i.e., a Malicious IP addresses DB supplied by ProofPoint) if you want to prevent attacks towards your public-facing workloads.



Aviatrix PSF Deployment Workflow (part.1)

To deploy a Public Subnet Filtering Gateway:

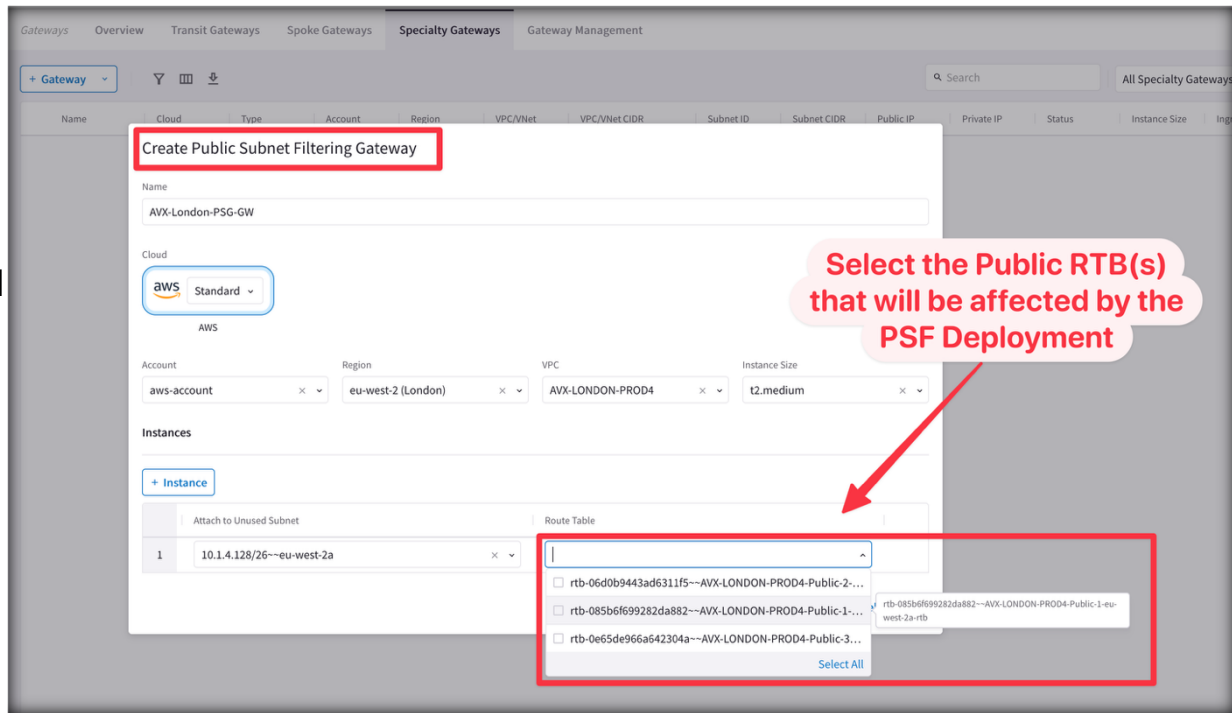
1. In CoPilot, navigate to **Cloud Fabric** > **Gateways** > **Speciality Gateways** tab.
2. Click **+Gateway** and select **Public Subnet Filtering Gateway**.



Aviatrix PSF Deployment Workflow (part.2)

3. Fill up the relevant fields with the required parameters.
4. Select the Public RTB that will get its default route affected (i.e. pointing to the PSF, instead of the IGW)

After the Public Subnet Filtering Gateway is deployed, **Ingress traffic** from IGW is routed to the gateway in a “pass through” manner. **Egress traffic** from instances in the protected public subnets is routed to the PSF gateway in a pass through manner.



The screenshot shows the 'Create Public Subnet Filtering Gateway' form in the Aviatrix console. The form is titled 'Create Public Subnet Filtering Gateway' and is located under the 'Specialty Gateways' tab. The form includes the following fields:

- Name:** AVX-London-PSG-GW
- Cloud:** AWS (Standard)
- Account:** aws-account
- Region:** eu-west-2 (London)
- VPC:** AVX-LONDON-PROD4
- Instance Size:** t2.medium
- Instances:** + Instance
- Route Table:** A dropdown menu is open, showing a list of route tables. The list includes:
 - rtb-06d0b9443ad6311f5--AVX-LONDON-PROD4-Public-2...
 - rtb-085b6f699282da882--AVX-LONDON-PROD4-Public-1...
 - rtb-0e65de966a642304a--AVX-LONDON-PROD4-Public-3...

A red box highlights the 'Route Table' dropdown menu, and a red arrow points to it with the text: "Select the Public RTB(s) that will be affected by the PSF Deployment".

Enforcement on PSF

The Enforcement of DCF (Distributed Cloud Firewall) rules on the PSF Gateway is *enabled* by default.

- CAVEAT: Keep this feature enabled to allow the AVX Controller to push DCF rules to this standalone gateway.

Enforcement on PSF Gateways

Control the application of Distributed Cloud Firewall Policy on PSF Gateways.

Status

✓ Enabled

Disable



Lab 5 – Aviatrix Cloud Firewall